

Acadaimy Cheat Sheet #3



1. Broadly, Speech Recognition allows for the

2. Examples of Speech Recognition software:

a. _____

b. _____

3. Speech Recognition works by:

a. Breaking down audio into _____

b. Converting these sounds into _____

c. Using algorithms and models to find _____

4. Speech Recognition Steps:

a. ADC translates sound waves to digital data

i. When people speak, they create _____

ii. The _____ transforms this sound wave into

iii. The ADC also _____

and _____

iv. It then separates the data into _____,

which the _____

b. Spectrogram analyses frequency of sounds

i. On the X-Axis, there is _____, and the Y-Axis plots the

ii. All words are made up of

iii. Vowel sounds have different

c. Machine detects phonemes and connects them to dictionary words

i. Phonemes are _____

ii. Computer runs phonemes through _____

I. Compares them to words in a _____

iii. Human language isn't simple because _____

d. Models examine complex patterns

i. Models such as the _____ allow the computer to understand the intricacies of our human language

ii. After Speech Recognition, _____ allows the machine to understand context, meaning, and purpose of our words

5. Smart Speaker Case Study – Alexa:

a. Trigger Word Detection

i. Input: _____

ii. Output: _____

b. Speech Recognition

i. Input: _____

ii. Output: _____

c. Intent Recognition:

i. Input: _____

ii. Output: _____

d. Execute Command

i. Speech Synthesis:

